ALLIANCE UNIVERSITY
Alliance College of Engineering and Design

Student-Centred and Industry-friendly
OBCFC Course Structure

PART B – Micro–Project Report

Course Code: E1CSA25202

Course Title: Programming in python

Name of Course Incharge: Deepak Raj.T

Payroll System for Salary Calculation Based on Wages, Working Hours, Overtime, and Deduction

1.0 Rationale

- In modern organizations, payroll management plays a crucial role in maintaining employee satisfaction and ensuring financial accuracy. Manual payroll calculations are time-consuming, error-prone, and inefficient, especially when dealing with multiple employees and complex salary structures. This micro-project aims to design and implement a payroll system that automates salary computation using parameters such as hourly wages, working hours, overtime pay, and tax deductions.

- The system ensures accuracy, transparency, and consistency in salary calculations. It also minimizes human intervention, thereby reducing calculation errors and saving time. This project is particularly important as it reflects real-world business applications and demonstrates how computational techniques can be applied to solve practical problems. Additionally, it enhances understanding of financial computations and strengthens programming and analytical skills.

2.0 Competency of the Course

This micro-project helps in achieving the following competencies:

- Ability to design structured and modular programs
- Understanding real-time application development
- Applying mathematical formulas in programming
- Developing debugging and testing skills
- Enhancing logical reasoning and algorithmic thinking
- Learning teamwork and documentation skills

3.0 Course Outcomes Addressed

- CO1: Apply programming concepts to solve real-world problems
- CO2: Design algorithms for computational tasks
- CO3: Implement structured programs using appropriate logic
- CO4: Analyze and interpret output data effectively
- CO5: Work collaboratively in teams and document the process effectively

4.0 Literature Review

- Payroll systems have become a fundamental part of organizational management systems. Traditionally, payroll was handled manually using registers and spreadsheets, which often led to errors and inefficiencies. With the advancement of technology, automated payroll systems have been introduced to streamline the process.
- Modern payroll systems integrate various features such as employee data management, attendance tracking, tax calculation, and payslip generation. These systems use predefined formulas and rules to calculate salaries accurately. Concepts such as gross salary, net salary, overtime pay, and tax deductions are essential components of systems.

- Several studies highlight that automation in payroll processing reduces errors by up to 90% and significantly improves efficiency. Programming languages like Python, Java, and C are commonly used to develop payroll systems due to their flexibility and computational capabilities.
- This project is based on fundamental payroll concepts and implements a simplified system using programming logic. It focuses on core calculations rather than advanced features like database integration or cloud deployment. The system uses conditional statements, arithmetic operations, and structured programming techniques to compute salary components**.**

5.0 Actual Methodology Followed

Step 1: Problem Identification

The goal was to create a system that calculates employee salary based on:

- Working hours
- Hourly wage
- Overtime hours
- Tax percentage

Step 2: Input Parameters

- Employee Name
- Hours Worked (H)
- Wage per Hour (W)
- Overtime Hours (OT)
- Tax Percentage (T%)

Step 3: Mathematical Model

- Basic Salary = H × W
- Overtime Pay = OT × (1.5 × W)
- Gross Salary = Basic Salary + Overtime Pay
- Tax Deduction = (T/100) × Gross Salary
- Net Salary = Gross Salary – Tax

Step 4: Algorithm

- Start
- Read employee details
- Calculate basic salary
- Calculate overtime pay
- Compute gross salary
- Apply tax deduction
- Calculate net salary
- Display results
- Stop

Step 5: Flowchart

Start → Input → Process (calculations) → Output → End

Step 6: Implementation:

```python
name = input("Enter Employee Name: ")
hours = float(input("Enter Hours Worked: "))
wage = float(input("Enter Wage per Hour: "))
overtime = float(input("Enter Overtime Hours: "))
tax = float(input("Enter Tax Percentage: "))

basic_salary = hours * wage
overtime_pay = overtime * (1.5 * wage)
gross_salary = basic_salary + overtime_pay
tax_deduction = (tax / 100) * gross_salary
net_salary = gross_salary - tax_deduction

print("\n--- Salary Details ---")
print("Employee:", name)
print("Basic Salary:", basic_salary)
print("Overtime Pay:", overtime_pay)
print("Gross Salary:", gross_salary)
print("Tax Deduction:", tax_deduction)
print("Net Salary:", net_salary)
```

Step 7: Testing

Test Case	Input	Output
Case 1	Normal values	Correct salary
Case 2	Zero overtime	No overtime pay
Case 3	High tax Reduced	net salary

6.0 Actual Resources Used

S. No.	Name of Resource/material	Specifications	Qty	Remarks
1	Laptop	8GB,i5Processor	1	Development
2	Python	Python 3x	1	Coding

7.0 Outputs of the Micro–project

System Features:

- Accepts user input
- Calculates salary instantly
- Displays structured output

Sample Output:

```
Enter Employee Name: Ravi
Enter Hours Worked: 45
Enter Wage per Hour: 150
Enter Overtime Hours: 5
Enter Tax Percentage: 12

--- Salary Details ---
Employee: Ravi
Basic Salary: 6750.0
Overtime Pay: 1125.0
Gross Salary: 7875.0
Tax Deduction: 945.0
Net Salary: 6930.0
```

8.0 Benefits of this Micro–project

- Reduces manual calculation errors
- Saves time and effort
- Improves understanding of salary structures
- Enhances programming and logical skills
- Useful for real-world applications
- Can be extended into full software

9.0 Area of Future Improvement

- Add GUI (Graphical User Interface)
- Store employee data using database (MySQL)
- Generate automated payslips
- Add multi-employee support
- Implement tax slabs instead of fixed percentage
- Develop web or mobile application version

10.0 References

1. K. Sharma, Computer Fundamentals and Programming, XYZ Publications, 2021.
2. Ram, Introduction to Python Programming, McGraw Hill, 2020.
3. “Payroll System Concepts,” GeeksforGeeks.
4. “Salary Calculation Techniques,” Tutorialspoint.

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